

MD. RAKIB HOSSAIN

Data Science and Machine Learning | Full-Stack Developer

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PROFESSIONAL SUMMARY

Computer Science and Engineering student who takes machine-learning systems end-to-end: dataset curation, exploratory analysis, feature engineering, model calibration, and deployment, and ships the applications around them. A deployed ML risk service, a live e-commerce platform with a recommendation engine mined from real order history, and a published hardware study. Uses AI-assisted development with Claude Code daily. Three papers accepted, including IEEE COMPAS 2026.

TECHNICAL SKILLS

Languages: Python, SQL, C++, TypeScript/JavaScript

Data Science & ML: pandas, NumPy, scikit-learn, CatBoost, PyTorch, YOLO, OpenCV, SHAP

Methods: Exploratory data analysis, feature engineering, association-rule mining, model calibration, leakage auditing, benchmarking, experiment design

Backend: NestJS, Node.js, REST APIs, Prisma, PostgreSQL

Frontend: React, Next.js, Tailwind CSS, HTML/CSS

Tools: Git/GitHub, VS Code, Jupyter, Roboflow, Claude Code, Antigravity, Canva, Microsoft 365, Vercel, Render

PROJECTS AND PUBLICATIONS

RideGuard - Calibrated and Auditable ML Risk Framework | Python, CatBoost, scikit-learn, SHAP | [Research](#) | [App Code](#) | [Live](#)

- A live web app scoring motorcycle route risk in Dhaka, with a colour-coded risk map and real-time warnings for phone handling, speeding, and accident hotspots.
- Predicts crash risk at 0.973 macro-F1, reporting calibrated confidence and explaining each prediction with SHAP attributions rather than acting as a black box.
- A self-audit layer traced the headline score to target leakage - a label the data gave away - which I published as the central finding rather than the accuracy number. Accepted at IEEE COMPAS 2026 (first author).

Carbon Market Simulation - CO2 Forecasting and Agent-Based Modeling | Python, CatBoost, Mesa, scikit-learn

- Benchmarked Ridge, XGBoost, LightGBM, and CatBoost on a 17-country CO2 panel; CatBoost won at 10.95% test MAPE.
- Fed forecasts into a Mesa agent-based model of 48 firms; linking markets cut costs in 55-62% of 100 runs. Third author.

DhakaNight - Night-Time Object Detection Dataset and Benchmark | Python, PyTorch, YOLO, OpenCV, Roboflow | [Code](#)

- Detects vehicles and pedestrians on Dhaka streets after dark, where standard detectors fail. Accepted at ICCA 2026 (first author).
- Curated and labelled a 2,300-image night dataset, then benchmarked CLAHE, Gamma, Zero-DCE, and RetinexFormer against a YOLOv8 baseline under a controlled protocol - none improved detection, and scaling the detector beat every pipeline.

PulseStone - Handheld Anxiety-Relief Device | ESP32-C3, Arduino C++, Python | [Code](#)

- A pocket device that starts a calming breathing exercise with one squeeze, guiding the user with light, vibration, and screen cues.
- Designed and analysed a controlled 15-participant, 150-trial study: users started in 0.52 s versus 7.49 s on a phone app - 14.5x faster - and all 15 preferred it. Accepted at ICCA 2026 (first author).

Drikon - Full-Stack E-Commerce Platform with Recommendation Engine | Next.js 15, React 19, NestJS 11, PostgreSQL, Prisma | [Live](#) | [Code](#)

- Implemented the Apriori association-rule algorithm from scratch - level-wise itemset mining with candidate pruning, scored by support, confidence, and lift - to power "frequently bought together" and personalised recommendations from real order history.
- Built 17 backend modules - products, orders, reviews, coupons, flash sales, wishlists, banners, settings - with voice search, filtering, comparison, and secure accounts on the storefront.
- Localised in Bengali and English, with branding, theme, and homepage content editable from the admin panel without a redeploy.

LEADERSHIP AND ACTIVITIES

Founder and Team Lead, "Khepa Chakka" Robotics Team | 2023 - Present

- Founded and lead a competitive robotics team building four classes of robots: line-following, soccer, race, and battle.
- Coordinate members across mechanical design, electronics, and software, holding timelines against competition deadlines.

Active Member, Engineering Students' Association of Bangladesh (ESAB), AIUB | 2023 - Present

- Collaborate with faculty on research and development; participate in seminars and workshops on emerging technologies.

AWARDS AND HONORS

- Winner, Cozmo Clench - Techfest, IIT Bombay (Bangladesh Zonal), 2024, one of Asia's largest science and technology festivals. Champion, Robot Soccer - AIUB CS Fest, 2024.
- National Robotics Championship, 2025: 1st Runner-up, Nano Battle Bot; 2nd Runner-up, Robo Race. 1st Runner-up, Robo Race - Technoxian Bangladesh National Round, 2024. 1st Runner-up, Robot Soccer - AIUB Robotic Crew (ARC).
- Competitor, BUET Autofest 2024 (Soccer Bot and Robo Race) and numerous national and inter-university robotics competitions.

EDUCATION

B.Sc. Computer Science and Engineering, American International University-Bangladesh (AIUB) | CGPA 3.30/4.00 | Expected 2026
Higher Secondary Certificate (HSC), Dhaka Imperial College, 2021 | Secondary School Certificate (SSC), Dhaka Collegiate School, 2019
Languages: Bangla (Native), English (Intermediate)